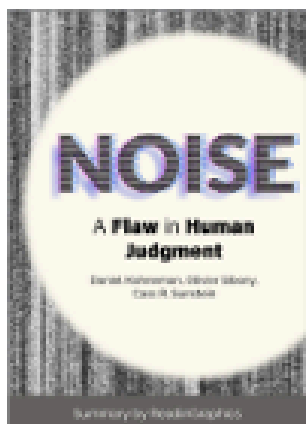




NOISE

A Flaw in Human Judgment

Daniel Kahneman, Olivier Sibony, Cass R. Sunstein



The Big “So What”

This book explores the concept of noise—an error of judgment that affects human decisions. The authors explain what is noise, how it affects decisions at all levels, and how we can mitigate noise to improve the efficiency, consistency and fairness of our decisions and systems. The ideas are relevant for decision-makers in all walks of life (e.g. executives, professionals, policy-makers), and anyone who's interested in learning about psychology, behavioral economics, and organizational behavior.



What is Noise and Why It Matters



Human judgments (e.g. forecasts, performance evaluations) are inconsistent and inaccurate. Given the same context and information, different people will make different judgments or decisions. Such inconsistencies come from noise and bias.

- Noise refers to the *random variability and inconsistencies* in judgment which lead people to evaluate the same information differently. It includes:
 - (i) In-person variability which occurs when the same person judges something differently under different circumstances.
 - (ii) Between-person variability which occurs when different people judge the same case differently.

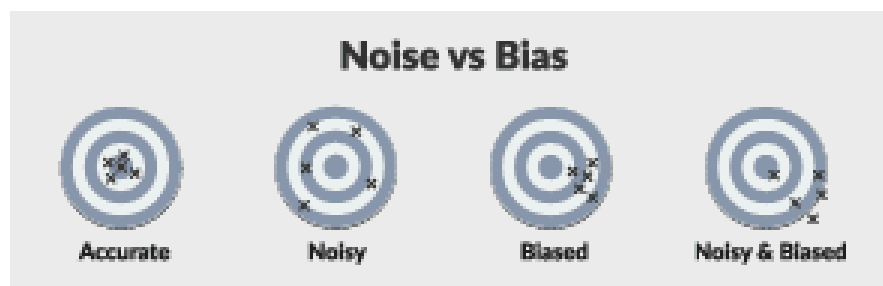
For example, 5 managers may evaluate the same business proposal differently. The same manager may even evaluate the same proposal differently on different days depending on his/her mood or focus.

KEY QUOTES

“System noise [is] unwanted variability in judgments that should ideally be identical.”

“We do not always produce identical judgments when faced with the same facts on two occasions.”

- Biases refer to the systematic errors in thinking that consistently skew judgments in a certain direction. For example, all 5 managers in a company may evaluate a business proposal too optimistically because of an optimism bias.



✖ Both noise and bias contribute to flawed decision-making, yet relatively little attention has been paid to noise compared to biases.

- In this book, the authors show how noise can lead to inconsistencies, unfairness, and inefficiency. They also recommend solutions for reducing noise and improving the quality of our decisions. In this summary, we'll explain noise and its implications, before moving into solutions for noise.



Understanding Noise



Noise Exists Everywhere



Noise exists in any situation that involves human judgments—from legal systems to medicine, financial markets, and hiring practices. For example, we expect the criminal justice system to be fair and consistent, when it's actually very noisy. A criminal can get a drastically different sentence depending who's the judge presiding over the case, or even the judge's mood that day. In the USA, the prison term for extortion ranges from 3-20 years, while the term for fraud ranges from 8.5 years to life imprisonment.



Noise exists even if it can't be detected or measured.

- Recurrent decisions are made repeatedly, and you can identify the extent/impact of noise through a "noise audit". For example, underwriters quote insurance premiums based on their clients' health situation. To assess the inconsistencies in quotations, an insurance company assigned 5 realistic

KEY QUOTES

"Wherever you look at human judgments, you are likely to find noise. To improve the quality of our judgments, we need to overcome noise as well as bias."

"Wherever there is judgment, there is noise—and more of it than you think."

"Our inability to observe variability [does] not make the decision less noisy."

client cases to different groups of underwriters. They found a median difference of 55%, which meant that the same client may be quoted \$9,500 or \$16,700 depending on which underwriter was assigned to the case. The variations were estimated to cost the company hundreds of millions of dollars. If the quotes were too high, the company could lose a potential client. If the quotes were too low, the company could lose money from large amounts of insurance claims.

- Singular decisions are one-off decisions for unique problems. We can't measure noise in such cases, e.g. you cannot compare 2 presidents' decisions to go to war because the circumstances are different. Yet, the same factors that cause noise in recurrent decisions also exist in singular decisions. So, it stands to reason that noise exists in singular decisions even if you can't measure it. The strategies that reduce noise in recurrent decisions should also improve the quality of singular decisions.



System noise—the unwanted variability within a system of judgments—is a problem because the goal of judgments is to be accurate, not to express personal preferences.

- Variability isn't always bad. For instance, it's fine for people to have different opinions about movies or wine. It's also helpful to explore diverse viewpoints for complex issues like economic forecasts or developing a new vaccine.
- However, system noise is always unwanted because it means inconsistencies in areas like criminal sentences, product service quality, or employee performance. You don't want a different sentence depending on the judge for a court case, nor a different diagnosis depending on the doctor you consult. Yet, noise is tolerated because we often don't even realize that it exists.



Evaluating Judgments and Measuring Noise



Judgments can be predictive or evaluative.

- A predictive judgment involves an objective value that you can compare your judgment with. But, not all predictive judgments are verifiable. You can verify a forecast of tomorrow's average temperature in London, but you can't

KEY QUOTES

"We cannot measure noise in a singular decision, but if we think counterfactually, we know for sure that noise is there."

"System noise is inconsistency, and inconsistency damages the credibility of the system."

verify a forecast of Earth's average temperature 1000 years from now. There are 2 measures typically used in predictions:

- (i) The *percent concordant (PC)* shows how well a prediction covaries with actual results. A perfect prediction has a PC of 100%.
- (ii) The *correlation coefficient (r)* is the percentage of shared determinants between 2 variables, and it ranges from -1 to 1. For example, if a physical trait is purely genetic, then we'll expect a 0.25 correlation in that trait between cousins with 25% common genes.

- An evaluative judgment (e.g. criminal judgments or ice-staking competitions) doesn't involve a value that you can validate. Good evaluative judgments should reflect the principles, standards, or values of an entire system, not individual beliefs.



Judgments can be evaluated based on their outcomes or processes.

- You can look at the outcome of a prediction to see how accurate it is. However, you can only evaluate a forecaster's accuracy over a large number of forecasts (not just a single forecast). Moreover, not all outcomes are verifiable. For example, if you didn't hire a candidate, you won't know if he/she could have succeeded in the role.
- Another way is to look at the process. Often, you can tell if a judgment is done properly just by looking at the process. This works for both verifiable and unverifiable judgments.



Every judgment involves some form of measurement. For example, judges measure the severity of a crime on a mental scale, while underwriters assign a dollar value to the risk of insuring someone. The best way to measure noise is by using standard deviation, which shows how much a judgment has deviated from the mean value. The Mean Squared Error (MSE) is widely used to evaluate the performance of predictive models and forecasts.

$$\text{Overall Error (MSE)} = \text{Bias}^2 + \text{Noise}^2$$

- A higher MSE means greater variability or inconsistency, whereas a lower MSE means greater accuracy or consistency. Reducing either noise or bias will reduce the overall error in judgment.

KEY QUOTES

"Reducing noise in predictive judgment is always useful, regardless of what you know about bias."

"Once noise is measured...it is often possible to reduce it."

- In noisy systems, errors add up; they don't cancel out one another. For example, an overpriced insurance premium doesn't compensate for an underpriced premium since they are for different clients.



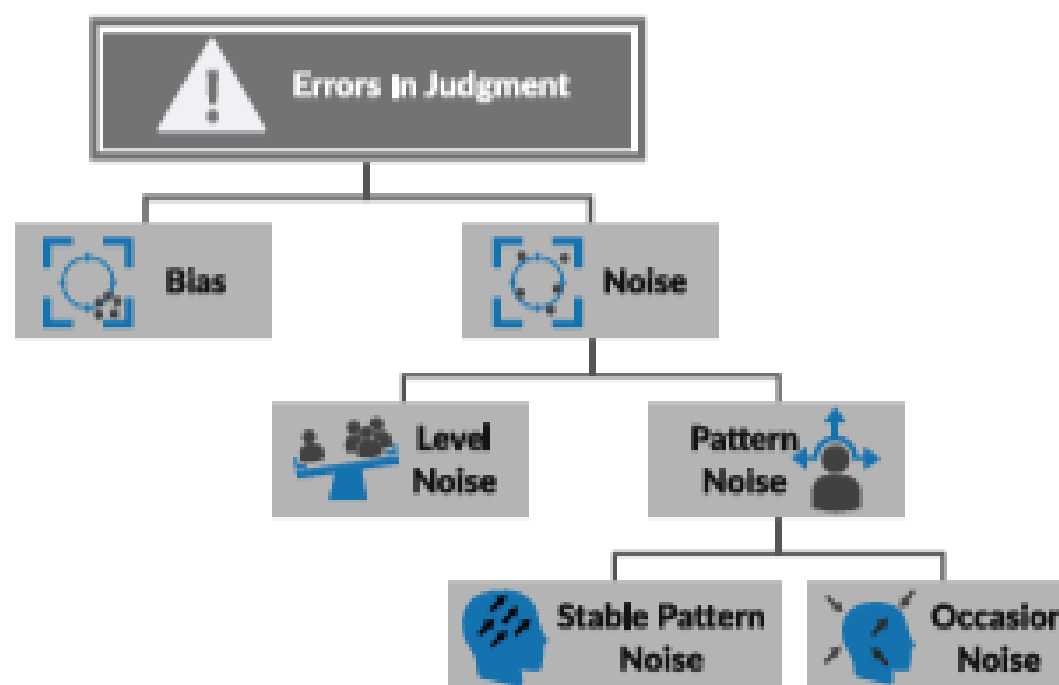
There are 2 main components of system noise:

$$\text{System Noise}^2 = \text{Level Noise}^2 + \text{Pattern Noise}^2$$

- Level noise occurs when a 1 person's judgments consistently differ from the average person's, e.g. some judges are generally more lenient than others.
- Pattern noise occurs when 1 person is consistently affected by specific factors, e.g. a judge is generally more lenient toward older criminals. It includes stable pattern noise and occasion noise.
 - (i) *Stable pattern noise* comes from our internal beliefs, biases, values, or knowledge/experiences. We may weigh some factors more heavily than others, or react positively/negatively to specific cues. Such patterns are relatively stable, though they're hard to predict.
 - (ii) *Occasion noise* refers to random errors that affect your judgment, such as your mood, weather, state of mind, or how the information is presented. For example, one study found that college admissions officers weighed academic credentials more heavily on cloudy days, and weighed non-academic factors more heavily on sunny days. Occasion noise can only be measured with large-enough samples that statistically show the effects of occasion-related factors.

*"No one is exactly like you...
your stable pattern errors are
unique to you."*

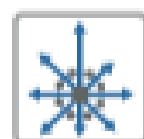
*"You are not always the same
person, and you are less
consistent over time than you
think."*



- In several studies, stable pattern noise was found to be the biggest component of noise, followed by level noise, then occasion noise. Such findings are far from conclusive, but they can help us to think about how to reduce noise.



Sources of Noise and Strategies for Noise Reduction



Where Does Noise Come From?

The components of noise (level noise, pattern noise, occasion noise) can be understood from a psychological perspective.



In the book *Thinking, Fast and Slow*, Daniel Kahneman explains how the brain uses mental shortcuts or heuristics to think rapidly and intuitively.

- However, this leads to psychological biases, including:
 - (i) *Substitution bias*: We unknowingly substitute a difficult question (e.g. "Do you think this company will be successful in 5 years?") with a simpler or more accession question (e.g. "Do I like this company's products?"). Or, you may answer the question "How satisfied are you with your life in general?" by considering your mood now and neglecting other factors.
 - (ii) *Conclusion bias*: We may bypass or distort evidence because we're already inclined toward a certain outcome/ conclusion.
 - (iii) *Halo effect*: Our judgment is influenced disproportionately by our initial impressions, e.g. you form a different impression if someone is described as "smart, persistent, sly, unprincipled" versus "sly, unprincipled, smart, persistent".

Psychological biases reduce our quality of judgments because they create *statistical bias* when the biases are widely shared, as well as *system noise* when individuals show different types/levels of biases.

- Another mental shortcut is "matching". For example, we may predict the chance of rain by looking at how cloudy the sky is,

KEY QUOTES

"Whether they cause statistical bias or noise...psychological biases always create error."

or predict how smart a kid is by looking at how much he/she reads. We also match intensities across different scales, e.g. comparing our love for music with the height of a skyscraper. Matching creates noise and errors because:

- (i) We make predictions using available info, even if the variables have *weak or non-existent correlations*. For example, how much a person reads is a poor predictor of how smart he/she is.
- (ii) We make *erroneous connections*, e.g. matching the traits "smart, meticulous, and unadventurous" to our stereotypical view of an accountant, when the same traits can also describe a sports athlete.
- (iii) We fail to consider *missing or conflicting cues* in complex problems, or we substitute them with simpler/accessible ones.
- Our brains can only discern up to 7 categories or match up to 7 values on an intensity scale (unless those categories are already well-established in advance). So, it's easier for us to make *comparative judgments* than absolute or categorical ones. For example, it's hard to rate 100 songs on a scale of 1-20. It's much easier to group them into 5 comparative categories from best to worst, then rate the songs within each category.



Noise also arises from the scales we use for making judgments. Different people use the scales differently:

- Words on the scale (e.g. "reasonable" or "excellent") may be *interpreted differently* by different people. We may also perceive the numerical values differently (e.g. 1 person may rate "good" as "3" while another rates it as "4").
- Judgments (like how severely a crime should be punished) may be driven by *emotions*, not facts.
- Judgments are the noisiest when people must *assign numbers arbitrarily*, e.g. when jury members must decide on the amount of damages for a crime without any context on what a fair amount may be. Once they conjure a figure, this creates an "anchor" that affects all their subsequent judgments.



Often, we feel overconfident about our judgments even when they're sub-optimal.

KEY QUOTES

"Comparative or relative judgments are more sensitive than categorical or absolute ones."

"Confidence is no guarantee of accuracy...and many confident predictions turn out to be wrong."

- Objective ignorance occurs when we're unaware that we lack information which affects the quality of our decisions. We feel confident about our judgments so long as our story seems "comprehensively coherent", i.e. all the details seem to fit (or we can ignore or justify things that don't fit) and there doesn't seem to be other viable explanations.
- When this happens, we get an internal sense of completion, and may even believe that others share our conclusions. Even if the judgment is wrong, the sense of false confidence is emotionally-rewarding.
- Most people naturally adopt causal thinking: we come up with stories about the cause-and-effect relationships between variables to explain what happened. This creates an *illusion of understanding*, i.e. we think we know what happened, and wrongly assume that we can predict similar events in the future. For example, we may conclude that someone succeeded because of his brilliant leadership, but fail to consider many other factors leading to his success.
- Noise becomes visible only when we adopt statistical thinking: to aggregate and analyze large amounts of data from multiple perspectives to work out patterns, correlations, differences, probabilities, etc. For example, to understand why a leader succeeded, you examine the predictors across many cases, then use statistical probability to identify possible root cause(s).



Causal thinking leads us to confuse causation and correlation.

- Correlation measures the degree to which changes in 1 variable are associated with changes in another variable. Causation refers to the cause-and-effect relationship between 2 variables, i.e. changes in 1 variable will cause changes in the other variable.
- Correlation doesn't imply causation, but causation always implies correlation. For example, there's a correlation between large feet and higher mathematical skills because both increase with age. However, you can't predict that someone will have higher mathematical skills because he has large feet, because there's no causal link between the two.
- You can make accurate predictions only if you understand the entire causal chain leading up to something. So long as there's

KEY QUOTES

"Objective ignorance sets a ceiling not only on our predictions but also on our understanding."

"The denial of ignorance is all the more tempting when ignorance is vast."

"Life is often more complex than the stories we like to tell about it."

"Causally, noise is nowhere; statistically, it is everywhere."

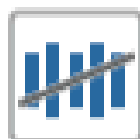
"Wherever there is causality, there is correlation."

"Causal thinking and the illusion of understanding the past contribute to over-confident predictions of the future."

an imperfect correlation (which is almost always the case), there's objective ignorance at play. Such errors in judgment are amplified for long-term predictions where there're exponentially more unknown factors involved.

🧠 Groups can amplify noise. Group decisions are affected by many seemingly-inconsequential factors, such as: seating arrangements, who speaks first, the speaker's body language, etc.

- Group judgments are so noisy due to several factors:
 - (i) *Information cascades*: people often follow others' opinions without evaluating the information independently. So, the initial opinion of 1 group member can set the tone for the entire group.
 - (ii) This is amplified by *social pressure*, when people agree with others to avoid seeming stupid or uncooperative.
 - (iii) *Group polarization* refers to the phenomenon where someone's opinion becomes more extreme after discussing it in a group. When juries deliberate their decision, they end up with a more extreme judgment than those held by individual jury members. If the median member was lenient, the entire jury became more lenient. When the median member was severe, the entire jury became more severe.
- Group settings are only beneficial when all the group members share independent answers. You can then aggregate and average the answers to reach a more accurate and less noisy judgment.



How to Reduce Noise

🔍 Start with a noise audit to understand the extent and impact of noise in your system. Simply assign the same problem to different people, compare their answers to measure the amount of variability in the judgments, and do a cost-benefit analysis to decide if the system noise is worth addressing.

🔍 Choose good judges with intelligence, experience/expertise and the right mindset. Above all else, ensure that the judges are *actively open-minded*, i.e. they actively seek out information to learn, test their hypotheses, and challenge their old beliefs.

KEY QUOTES

"Social influences are a problem because they reduce group diversity without diminishing the collective error."

"If group members are listening to one another, they will shift in the direction of the dominant tendency, rendering the group more unified, more confident, and more extreme."

"Aggregation can only reduce noise if judgments are truly independent."

"Good judgments depend on what you know, how well you think, and how you think."

KEY QUOTES

- (ii) You can also select a handful of the *best judges* based on the accuracy of their recent judgments, then take an average of their predictions. You can even add monetary incentives for the most accurate forecasts.
 - (iii) The *Delphi method* is a multi-round process that combines aggregation and social learning. Through a moderator, people anonymously share their estimates and the reasons for them, then adjust their estimates for each round until the numbers converge into an aggregate score.
 - (iv) The *estimate-talk-estimate method* simplifies the Delphi method into just 2 rounds. The aggregate is calculated by averaging the 2nd round of predictions.
- Improve group judgments by getting members to come up with independent answers before aggregating them.
 - Formulas—regardless of whether they are randomly-generated, modelled after people, or modelled after reality—will always outperform human experts in predictive judgments. Mechanical methods and algorithms have been found to be more accurate, faster and cheaper than human judgments, because they apply the same rules and weightage consistently to reduce noise.
 - Often, we make decisions intuitively after doing a quick mental calculation of the scores for each criteria. This is known as *clinical judgment*, and it's full of noise. For example, we may mentally assign different weightage to the same criteria at different times, and/or consider irrelevant factors unknowingly.
 - *Mechanical judgments* capture decision-making rules using statistical models or artificial intelligence. They may use multiple regression models (which assign optimal weights to specific predictors), equal weights models (which assign equal weights to all predictors), or frugal models (which use simple rules with a small set of strong, correlated predictors). They may also use artificial intelligence and machine learning to handle huge amounts of data, numerous predictors, and complex patterns beyond the capabilities of the human mind.

"Simple rules that are merely sensible typically do better than human judgment."

"Reducing noise mechanically increases the validity of predictive judgment."

"What AI does involves no magic and no understanding; it is mere pattern finding."



The mediating assessment protocol integrates many of the decision hygiene strategies above. It can be used/adapted by any organization to reduce noise in decisions. The key steps are:

- Break down the decision into a set of pre-determined criteria or mediating assessments.
 - (i) Get people to come up with a list of criteria independently before discussing them. Agree on the criteria and how to measure/assess them. Incorporate outside views where possible (e.g. getting a 2nd opinion by experts).
 - (ii) Ideally, develop a scale with a clear frame-of-reference: give clear descriptions and case-study videos for each rating, so raters have a clear baseline for comparative judgments.
- Use the same criteria/assessments to analyze recurring decisions.
 - (i) For each case, get people to review problems separately, then, use the estimate-talk-estimate method to share and discuss their inputs.
 - (ii) By now, the decision-makers all have detailed, relevant info. Calculate and share the average ratings for each criteria, then make a final decision as a team.



Noise Reduction in Real Life

The noise-reduction strategies above can be used to improve all types of judgments. Here are 3 examples of real-life applications.

 Forecasts (e.g. sales or project costs/durations) affect all sorts of organizational decisions. To get a more accurate forecast, you can: (i) select good judges who're strong individually and complement one another, (ii) train them to identify/avoid biases, and (iii) get them to make their forecasts independently before aggregating them.

 Medicine. There's a huge amount of noise in medical diagnoses and treatments due to factors like stress/fatigue, lack of skills, and a lack of clear guidelines/processes, etc. To reduce noise, you can:

- Use selection and training to increase skill levels;
- Aggregate judgments, e.g. getting a 2nd expert opinion;

- Use algorithms to help with diagnosis and treatment identification; and
- Develop diagnostic guidelines to reduce variations in human judgment. For example, the Apgar score uses 5 criteria (skin color, heart rate, reflexes, muscle tone, and respiratory effort) to help medical professionals assess if a newborn baby is in distress.



The employee hiring process is meant to help organizations select candidates who're likely to perform well in their roles. Yet, most companies use unstructured interviews which are practically useless in identifying good candidates. The solution is to switch to a more structured process:

- Break down the hiring decision into component parts (e.g. leadership, cultural fit, or role-specific knowledge) and focus only on data that measure those components.
- Give each rater the tools to independently assess and rate the candidates in each of the pre-determined components.
- Then, get a hiring committee to evaluate all the ratings along with additional data (e.g. test results, references) to make the final decision.



How Much Noise Should We Tolerate?



It's hard to enforce noise reduction strategies because most people don't even realize that noise exists. Others are concerned that such strategies may be rigid, unfair, or dehumanizing.

- In the 1980s, US congress members sought to address the disparity in criminal judgments by introducing a set of mandatory sentencing guidelines. This successfully reduced variability in criminal sentencing. Yet, people objected that the guidelines were too mechanical and didn't give the judges enough discretion to consider the nuances of each case. In 2005, the guidelines were finally changed from mandatory to advisory. The disparity in judges' sentences doubled thereafter.



There are 7 key objections to reducing or removing noise:

KEY QUOTES

"Unlike rules or algorithms, guidelines do not eliminate the need for judgment: the decision is not a straightforward computation."

"Our intuition will serve us much better once it is informed by the separate assessments we have asked for."

- It can be costly (and sometimes unviable) to reduce noise. That's why we need noise audits to learn the extent/impact of noise and run cost-benefit analyses.
- Noise-reduction strategies may create unwanted side-effects or errors. For example, if every forecaster makes the same prediction, the forecasts will be less noisy but also incorrect. People may focus too much on noise elimination to lose sight of other goals, principles, or values. Algorithms may also have built-in noise if (i) they use predictors correlated with traits like race or gender, or (ii) the source data itself is biased.
- Some level of noise must be tolerated if we were to let people preserve a sense of dignity and autonomy. Concepts like "mercy" and "empathy" are inherently subjective and noisy. It can seem dehumanizing if someone can't get a mortgage due to his credit rating, or can't get a job due to a drug offence 20 years ago.
- Rules that eliminate noise could also block new values that are vital for social/moral/political change. For example, our definition of women's role in society has changed over the years. This would not have been possible if our systems are governed by strict and unchanging rules.
- Rule-bound, noiseless systems may be more easily manipulated, compared to noisy systems where people can use their discretion to adapt to unexpected situations or system gaps.
- In some cases, noisy systems may be more effective as a deterrent for wrongdoings. Risk-averse people may shy away from a crime if they're unsure if it will lead to harsh punishments.
- Squashing noise could inhibit creativity and self-expression. These could lead to poor morale and performance, to generate even higher costs.



The authors believe that it's possible to address these concerns and reduce noise. The key is to adopt the right strategies, such as aggregating independent judgments or using shared scales grounded in outside views.



A related question is: should we regulate noise through rules, standards, or both?

KEY QUOTES

"It might be costly to remove noise—but the cost is often worth incurring."

"Algorithms could eliminate unwanted variability in judgment but also embed unacceptable bias."

"If we want people to feel that they have been treated with respect and dignity, we might have to tolerate some noise."

"A rule-bound system might eliminate noise, which is good, but it might also freeze existing norms and values, which is not so good."

"Our general conclusion is that even when the objections are given their due, noise reduction remains a worthy and even an urgent goal."

KEY QUOTES

- Rules like "Don't exceed 80 km/hour" or "Take 1 pill every morning and every night" are very specific and rigid. They remove discretion for the judges, allowing people to just follow the rules without much thought.
- Standards like "Drive at a safe speed" or "Take a pill whenever you need it" are more flexible. They allow the judges to apply discretion, but also makes it more effortful since the judges must decipher what words like "safe" or "necessary" mean in practice. In some case, guidelines like the Apgar scores may be specific, yet provide some room for discretion.
- To choose between rules and standards, you must consider the costs of decisions vs errors. As a rule of thumb, use rules if you need to minimize errors, and use standards if you need to retain some discretion.
 - (i) When there are many noisy decisions to be made, or when you can't trust the judges, it's better to have clear rules. For example, the law may say that only those aged ≥ 18 can buy liquor. This is to protect young people from the risks of alcohol consumption, even though someone who's 16 or 17 may handle alcohol more responsibly than someone who's 46 or 67. It's too messy to set broad standards and leave the discretion to alcohol merchants, so a rule becomes the logical solution, however imperfect it is.
 - (ii) However, if a rule may lead to disastrous mistakes (e.g. in matters of life and death), then a standard may be better to give the judges some discretion.

"Rules are meant to eliminate discretion by those who apply them; standards are meant to grant such discretion."

"Although accuracy is the goal, perfection in achieving this goal is never achieved even in scientific measurement, much less in judgment."



Conclusion and Other Details in the Book



Biases are widely discussed because they're more visible, and laws have even been enacted to correct/prevent unacceptable biases. On the other hand, noise is neglected because it's often invisible, even though it has at least as much impact as biases. To improve the effectiveness and fairness of our systems, it's crucial for organizations to start pay more attention to the costs and impact of noise.

"In many areas, the current level of noise is far too high. It is imposing high costs and producing terrible unfairness."